CHEM001x – Introductory Session

Warning: Today we will give you a LOT of information!

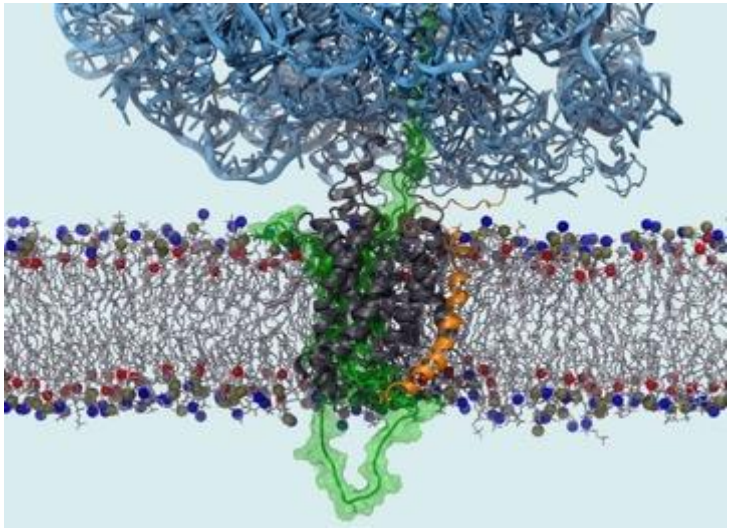
- Go to Menti.com: **3914 3803**

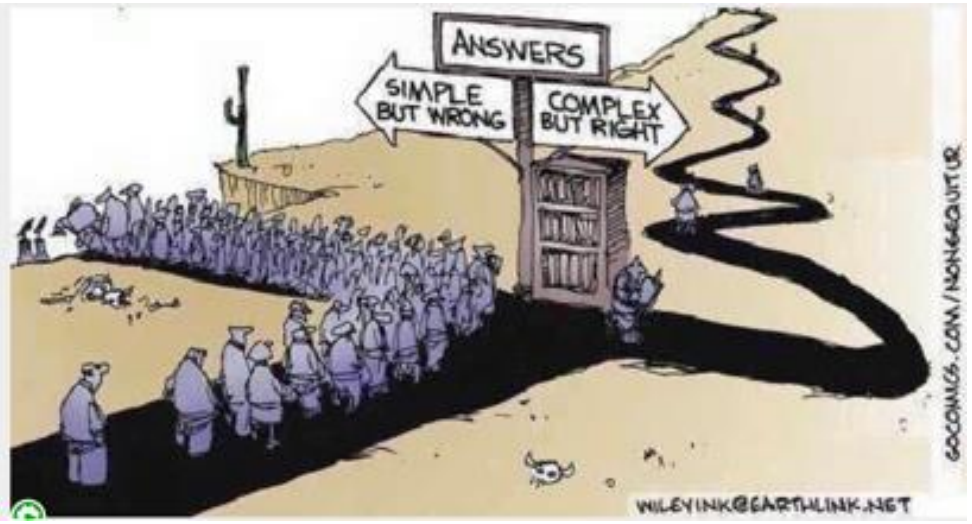


- While you wait – add a profile image to Moodle

CHEM001x (x = 0, 1)

Chemistry and Numeracy for Life Science Students





Email: chem001x@ucl.ac.uk (for personal admin issues only)

Any questions that does not only concern *you exclusively* should be posted to the Forum.

Check **Handouts**, **Forums,** and **Glossary** before sending email

What you can (and cannot) expect of us

- Our focus is on your learning.
- We will challenge you.
- We will answer questions on discussion fora within two working days (often much faster)
- But… we are not a 24/7 call centre.
- We will **NOT** answer questions about chemistry content via email – use the fora.
- We will be available in our student hours – It is best to drop us a line in advance to make an appointment.
- We will provide explanatory feedback to test questions we set.

What we expect of you

- We assume you are here to **learn** and not just get marks.
- You will **attend** ALL of our classes – lectures and workshops.
- You will **borrow, buy or Bibliu the textbook,** either new or secondhand, and **read** the relevant chapters – they are listed in handouts.
- You will submit **pre-qual answers** before workshops and submit feedback for your peers.
- You will interact in class workshops – **tap in when you arrive.**
- You will spend **around 3-4 hours/week** over the entire year working on the material from this course.
- You will **participate actively** in the Moodle discussion fora by posting questions, by answering questions from other students, contributing to the glossaries, etc. You will also attempt all questions in the revision quizzes, etc.
- You will post questions about course material on **discussion fora** rather than send emails to lecturers.

Recording Your Attendance

- You must record your attendance at lectures
- Download the SEATs Mobile app – to register your attendance:
  - In the first 20 minutes of the lecture
  - Open the SEAtS app and accept the prompt message requesting access to their Bluetooth and location
  - Your smartphone will find the beacon
  - After a few seconds the **Finding** sign will change from a blue to a green tick = success.

Try this now!

General Information

Prerequisite: Chemistry A-level (or equiv.)

CHEM0010 – 15 Credit

- OTZ (4 workshops)
- 5 sections (~25 Lectures)
- Interactive workshops (>9)
- Lab_14 (Part 1) (25 %)
- Workshop submission and peer assessment (10%)
- 2 hour online final exam (65 %)
  Structure:
  - Exam takes place online. Questions will overlap across sections and include practical material.

CHEM0011 – 30 Credit

- OTZ (4 workshops)
- 8 sections (~40 Lectures)
- Interactive workshops (>12)
- Lab_14 (Parts 1 & 2) (25 %)
- Workshop submission and peer assessment (10%)
- 2 hour final exam (65 %)
  Structure:
  - Exam takes place online. Questions will overlap across sections and include practical material.

Lecture Sections

- **OTZ**: Estimation and order of magnitude (Sella)
- Section **A**: Structure, Bonding, and the Self-Assembly of Biological Structures (Roffey)
- **Section B**: Seeing Bio-molecules – Spectroscopy and Quantification (Cassidy and Kiappes)
- **Section C**: Energy Flows and the Thermodynamics of Life (Cassidy)
- **Section D**: Concentrations, Potentials and Equilibria (Caruana)
- **Section E**:The Speed of Life – Chemical and Enzyme Kinetics (Riley)

CHEM0011 only:

- **Section H**: The Inorganic Chemistry of Life (Potts)
- **Section F**: The Bricks of Biological Architecture – Functional groups (Bucar)
- **Section G**: Macromolecules – the Spokes and Wheels of the Cell (Bucar)

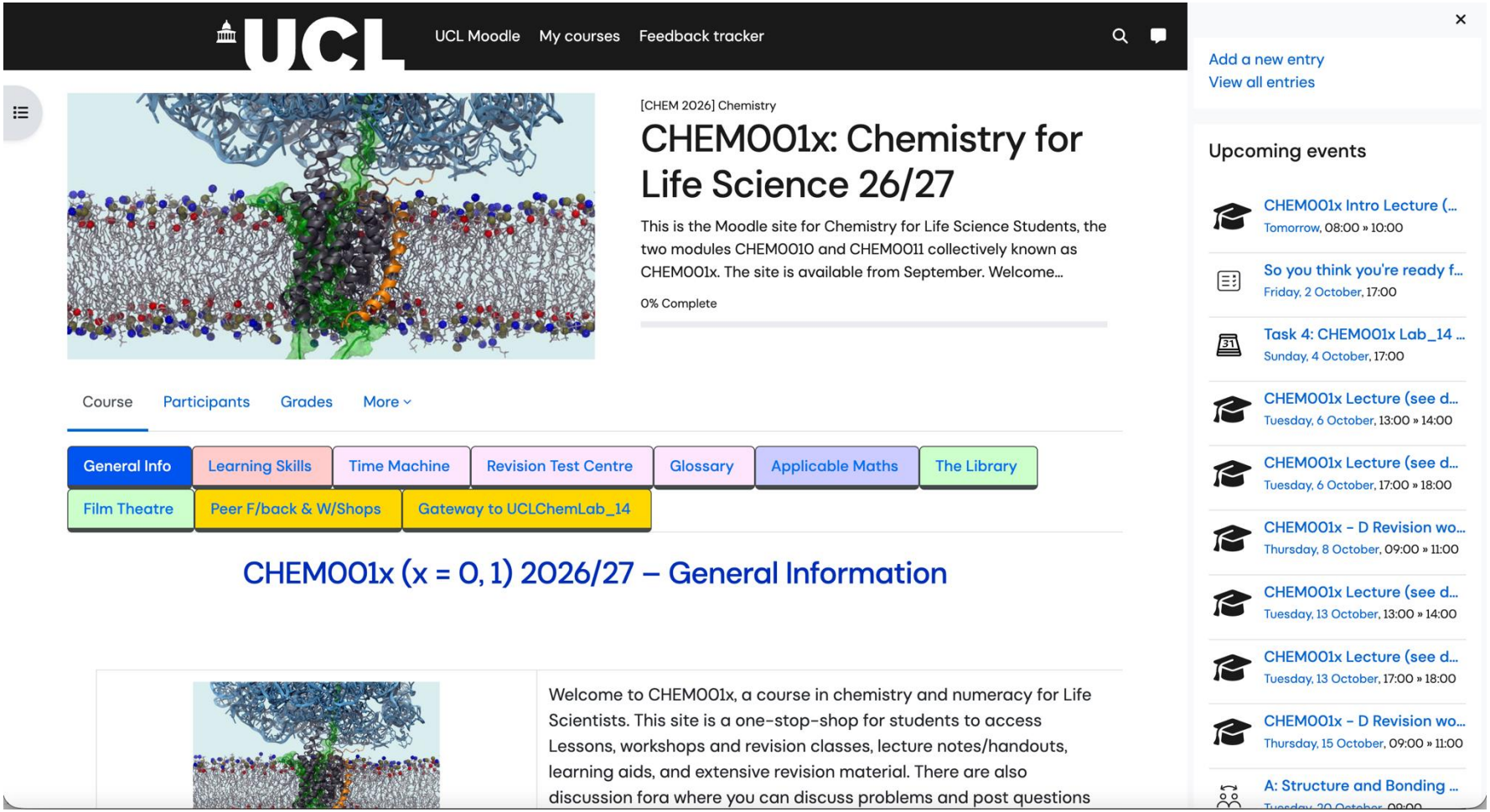
Moodle Virtual Learning Environment

https://moodle.ucl.ac.uk/course/view.php?id=61631

- **Lecture Handouts**
- **Course Calendar** – class times, upcoming events, deadlines etc.
- **Revision problems** and feedback

  "**One must learn by doing the thing; for though you think you know it, you have no certainty until you try.**" Sophocles
- **Course announcements**
- **Forums** for questions – tutors will not answer questions by email.
- **Interactive Glossary** – editable – share your own definitions, examples, ideas
- **Revision Self-Tests**
- **Additional resources**
  - **Self-study Lessons**
  - **Further reading**
  - **Links to media and web to watch**

Moodle


UCL
UCL Moodle
My courses
Feedback tracker
[CHEM 2026] Chemistry
CHEM001x: Chemistry for Life Science 26/27
This is the Moodle site for Chemistry for Life Science Students, the two modules CHEM0010 and CHEM0011 collectively known as CHEM001x. The site is available from September. Welcome...
0% Complete
Course
Participants
Grades
More
General Info
Learning Skills
Time Machine
Revision Test Centre
Glossary
Applicable Maths
The Library
Film Theatre
Peer F/back & W/Shops
Gateway to UCLChemLab_14
CHEM001x (x = 0, 1) 2026/27 – General Information
Welcome to CHEM001x, a course in chemistry and numeracy for Life Scientists. This site is a one-stop-shop for students to access Lessons, workshops and revision classes, lecture notes/handouts, learning aids, and extensive revision material. There are also discussion fora where you can discuss problems and post questions
Add a new entry
View all entries
Upcoming events
CHEM001x Intro Lecture (...
Tomorrow, 08:00 » 10:00
So you think you're ready f...
Friday, 2 October, 17:00
Task 4: CHEM001x Lab_14 ...
Sunday, 4 October, 17:00
CHEM001x Lecture (see d...
Tuesday, 6 October, 13:00 » 14:00
CHEM001x Lecture (see d...
Tuesday, 6 October, 17:00 » 18:00
CHEM001x - D Revision wo...
Thursday, 8 October, 09:00 » 11:00
CHEM001x Lecture (see d...
Tuesday, 13 October, 13:00 » 14:00
CHEM001x Lecture (see d...
Tuesday, 13 October, 17:00 » 18:00
CHEM001x - D Revision wo...
Thursday, 15 October, 09:00 » 11:00
A: Structure and Bonding ...

Colds, Flu, COVID and Biosecurity in class

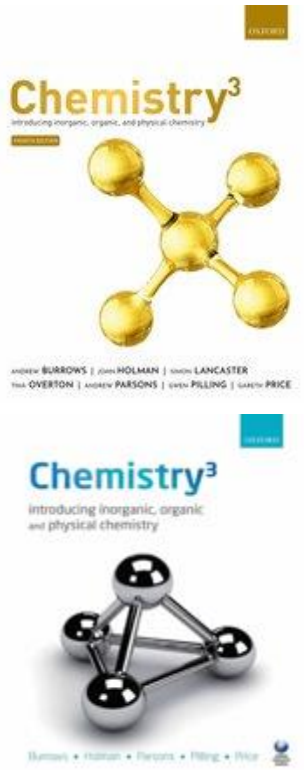
- COVID has not gone away
- Acute COVID could put you in bed for 1, 2, or even 3 weeks during term time (i.e. 5, 10, 15% of our teaching terms)
- Long COVID (which affects >17 million people in the EU) could affect your energy levels and cognitive abilities for several months.
- COVID is airborne.

- Masks are a smart idea. (**Ref**: *JAMA Netw Open.* 2022;5(8):e2225430)
- If you are ill, stay at home.
- Make sure you are vaccinated (measles, mumps, rubella, meningitis, HPV, flu and COVID).
- Register with a GP.

Ridgmount Practice
8 Ridgmount Street
London
WC1E 7AA

Ridgmount Practice has close UCL links & can provide EC forms if your health problems affect your studies

Recommended Textbook



Burrows Holman *et al.* "**Chemistry³**", OUP.

25% discount available from OUP:

https://global.oup.com/academic/product/chemistry-9780198733805?cc=gb&lang=en&

Use the code: WEBXSTU96

Or get current or previous editions from DMS Watson library or from older students.

Hard copies are better to read than online books.

Many alternatives are available from the DMS Watson Science Library:

- Atkins and Jones, "**Chemical Principles – The Quest for Insight**", Freeman
- McMurry and Fay "**Chemistry**", Pearson
- Housecroft & Constable - "**Chemistry**" Prentice Hall
- Hart, Craine & Hart "**Organic Chemistry – A Short Course**" Houghton & Mifflin
- Loudon "**Organic Chemistry**" Benjamin/Cummings.

Workshops (and peer assessment)

Throughout the term we will run *revision workshops* (interactive problem classes)

Peer feedback teams: Groups of 10 students from across the course.

- **You must submit pre-qualifying work.** Submission is worth 5%

- If you have submitted **you provide feedback** to three students in your team. Giving feedback is worth 5%.

Deadlines are FINAL and there will be NO extensions.

Workshop/Lecture Etiquette

- Classes start on the hour and end after 50 mins.
- Record your attendance
- Sit near the front – gives you better focus.
- If you're late, come in quietly and from the back.
- Sit with someone; you can solve problems together.
- Keep chatter down – very distracting for neighbours and for the lecturer
- Bring a pad of paper and have a calculator handy
- Bring an electronic device for Mentimeter
- Keep screens low as they are distracting for others.
- Ask questions either via Mentimeter or raised hands
- Spot mistakes! Then raise your hand and wait.

Revision Tests

- We will hold two tests during the year, right after the end of term to give you a chance to see the style of our final exam.
- Tests are **SYNOPTIC** and cover course material, practicals and maths/numeracy
- Tests will last 40 minutes (adjusted for SORAs) with ~ 30 questions.
- Questions are drawn in part from revision quizzes.
- If you have a timing clash we can rearrange things.
- The “grades” will give you a sense of what topics you are confident about and where you need to focus your attention.

Assessment: The Final Examination

- Takes place in May 2026
- The **date is not known** until late March 2026
- Further info on Final Exam tab (available in March)
- Exam format
  - Exam is on Moodle but takes place in an exam venue.
  - Closed book – a data/formula sheet will be provided in advance.
  - Typically 60ish questions
  - Question types as for our practice tests during term time plus:
    - Free text entry to explain something.
    - Draw sketch or diagram.
    - Upload a graph
  - Exam includes OTZ and material based on Lab_14 practicals.

#UCLChemLab_14

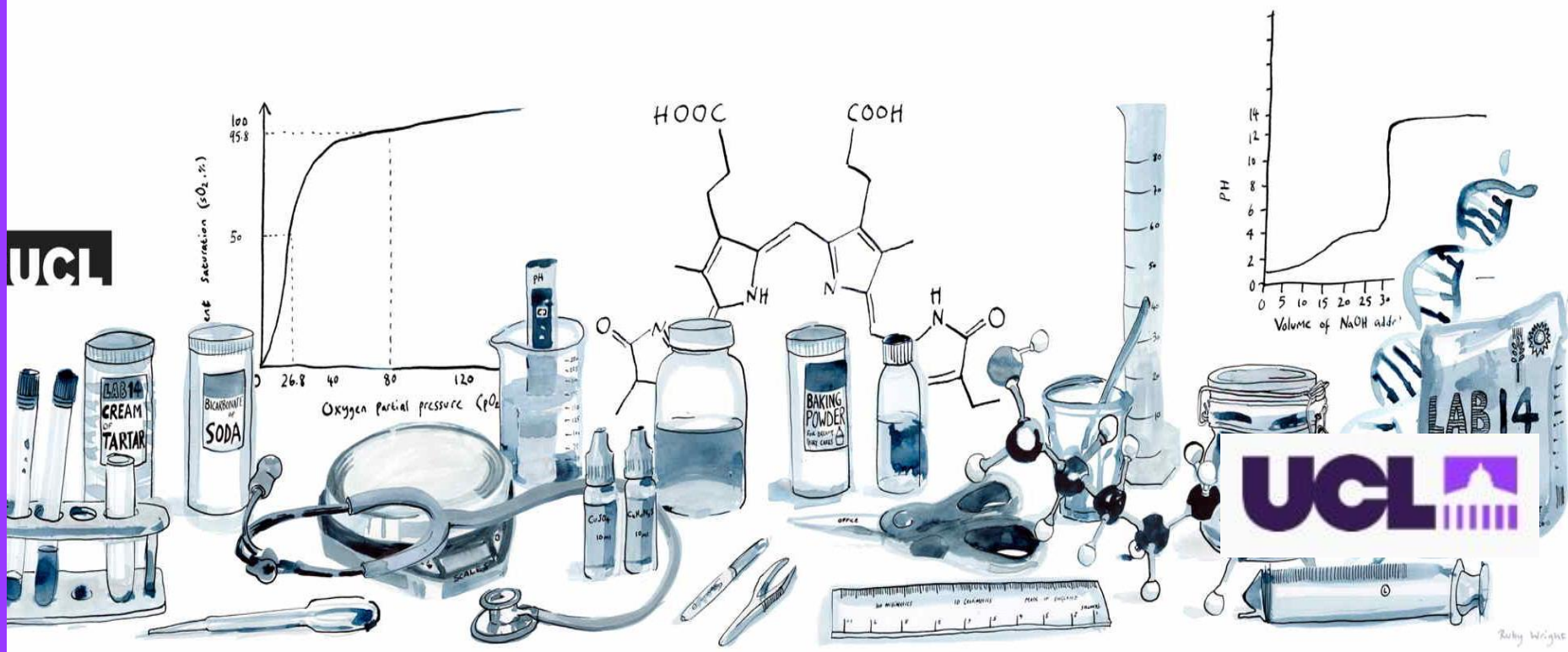
- This is an entirely new practical course that was designed from scratch in the summer of 2020.
- You have or will receive a kit containing lots of instruments
- You will also receive a Lab_14 Handbook that will give you instructions.
- All activities are supported through the Lab_14 Moodle site: https://moodle.ucl.ac.uk/course/view.php?id=46867
- As you do the exercises and activities you will document them through data uploads and photographs.
- Students are expected to complete the parts of the Handbook specified in the Lab_14 FAQ but you can do more if you like.
- You will get a mark based on the fraction of the number of activities you complete.
- You will be examined on @UCLChemLab_14 material.

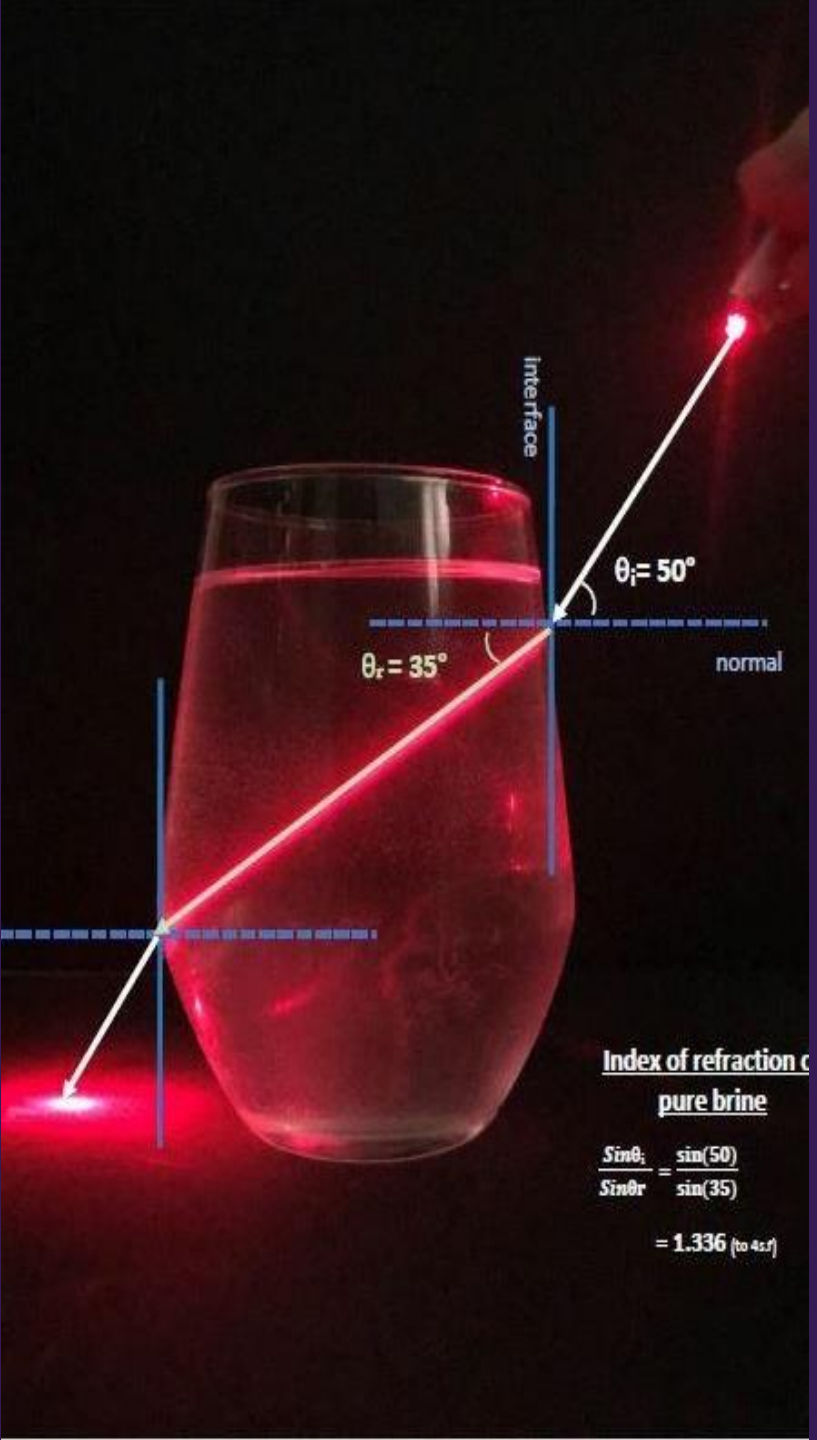
Your To Do List for 01/10/2026

1. Put an image in your profile.
2. Download the course synopsis and read it.
3. Check out/export the Calendar to YOUR calendar
4. Explore CHEM001x site – more will open week by week
5. Book a slot and collect your Lab_14 kit
6. Explore the Lab_14 website
7. Make a study schedule for the coming weeks.
8. Set up a small study group for this course.
9. Download the handout and from Saturday get cracking on Section A!

CHEM001x – Lab 14 Introduction

Dr Matthew Blunt





Welcome to Lab 14 2026/2027

Lab 14 is the practical component of the CHEM001x course.

You will be doing practicals that range from performing calorimetry with a kettle, to making your own pH indicator from a cabbage - and all from the comfort of your own home!

Lab 14 – What is it and why are we doing it?


thesis

Practical science at home in a pandemic world

- Lab 14 - developed during Covid when no one could get into a lab!
- New approach to the **chemistry set** – based on instruments and measurements, rather than "chemicals".
- Practicals are open ended – curiosity and collaboration are encouraged (as is cake).





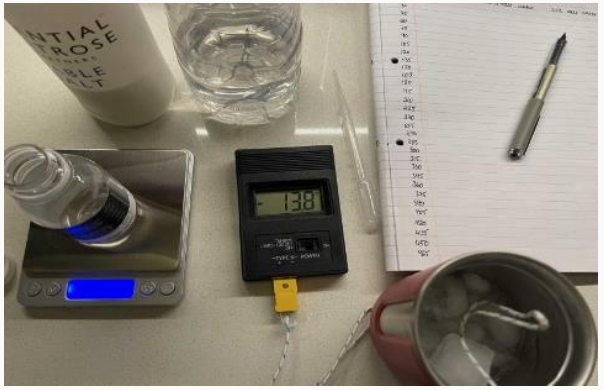
Lab 14 – The kit



- At some point over this week or next, you will book a slot to come and collect your Lab 14 lab kit (details on how to do this later).
- The lab kit contains a range of equipment including a digital thermometer, multi-meter, digital pH meter and mechanical pipette.
- Try and look after it - at the end of the year we would like it back!

Lab 14 – Common questions

- **Is Lab 14 compulsory?** – Yes, Lab 14 is a compulsory part of CHEM0010 and CHEM0011 – essential for course progression!
- **Can we work together?** – Yes, In fact we encourage it! Sharing knowledge and experience is part of science. Each student should do their own measurements and submit their own data.



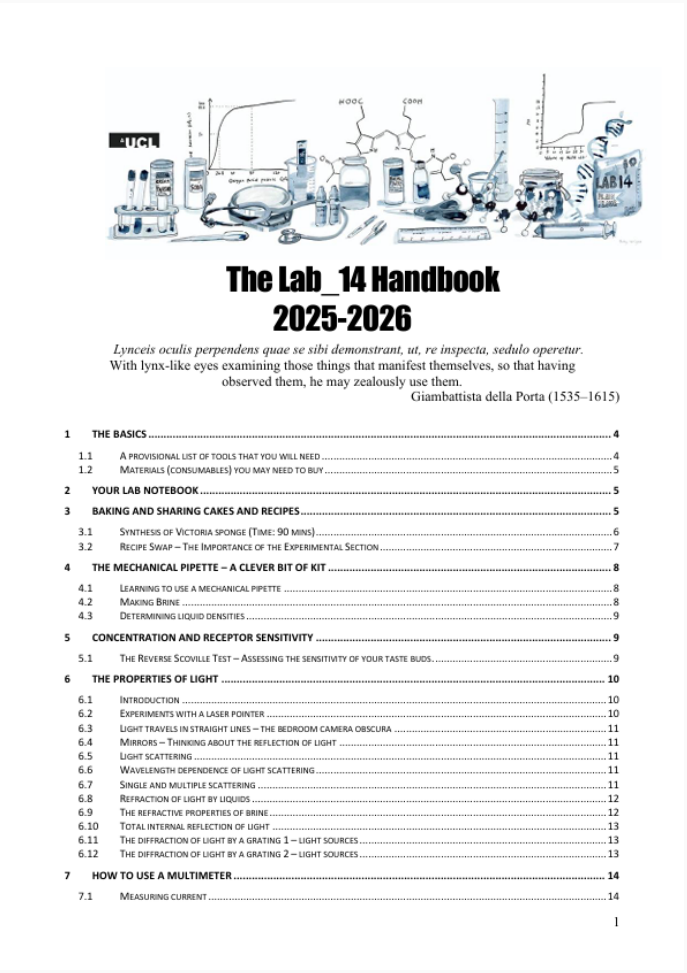


Lab 14 – What will you be doing.

- The course consists of a range of practicals; each should take 30 to 60 mins.
- Submissions are all made via the Lab 14 Moodle page.
- Expectation dates rather than deadlines - try to submit 1 practical per week.

- **CHEM0010 students**
- → minimum of Chapters 1 – 16

- **CHEM0011 students**
- → minimum of Chapters 1 – 20



The Lab_14 Handbook
2025-2026

Lynceis oculis perpendens quae se sibi demonstrant, ut, re inspecta, sedulo operetur.
With lynx-like eyes examining those things that manifest themselves, so that having observed them, he may zealously use them.
Giambattista della Porta (1535–1615)

1 THE BASICS 4
1.1 A provisional list of tools that you will need 4
1.2 Materials (consumables) you may need to buy 5
2 YOUR LAB NOTEBOOK 5
3 BAKING AND SHARING CAKES AND RECIPES 5
3.1 Synthesis of Victoria sponge (Time: 90 mins) 6
3.2 Recipe Swap – The Importance of the Experimental Section 7
4 THE MECHANICAL PIPETTE – A CLEVER BIT OF KIT 8
4.1 Learning to use a mechanical pipette 8
4.2 Making Brine 8
4.3 Determining liquid densities 9
5 CONCENTRATION AND RECEPTOR SENSITIVITY 9
5.1 The Reverse Scoville Test – Assessing the sensitivity of your taste buds 9
6 THE PROPERTIES OF LIGHT 10
6.1 Introduction 10
6.2 Experiments with a laser pointer 10
6.3 Light travels in straight lines – the bedroom camera obscura 11
6.4 Mirrors – Thinking about the reflection of light 11
6.5 Light scattering 11
6.6 Wavelength dependence of light scattering 11
6.7 Single and multiple scattering 11
6.8 Refraction of light by liquids 12
6.9 The refractive properties of brine 12
6.10 Total internal reflection of light 13
6.11 The diffraction of light by a grating 1 – light sources 13
6.12 The diffraction of light by a grating 2 – light sources 13
7 HOW TO USE A MULTIMETER 14
7.1 Measuring current 14

1

Lab 14 – How will I be assessed?

- **We want to assess how you engage with the practicals** – not what numbers you get fr
- Subm
- **Mar**
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- See
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The aim with Lab 14 is that you get to engage with the practicals and use your curiosity to learn more about the science covered in the CHEM001x course.

And above all, to have some fun!





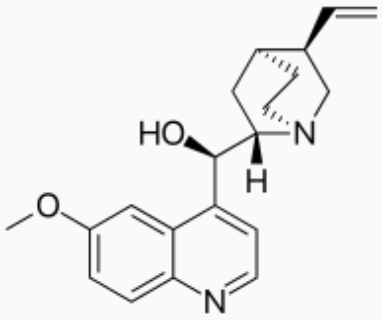
Baking a Cake: *Recipes and Experimental Methods*



- Gets you thinking about how to clearly describe experimental methods in a way other people can easily follow.
- Limiting reagents and the role they play in determining the scale of your reactions.
- Most importantly, you get to eat cake!

Serial Dilution – *The Inverse Scoville Test*

- Make serial dilutions of quinine.
- Accurately use a mechanical pipette and calculate how the correct volumes for diluting a solution.
- Taste them in reverse order to establish your receptor threshold.





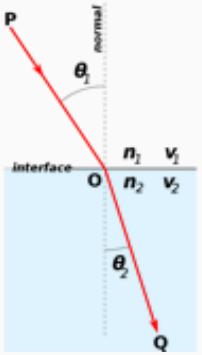
Properties of light – *Using a Laser Pointer*









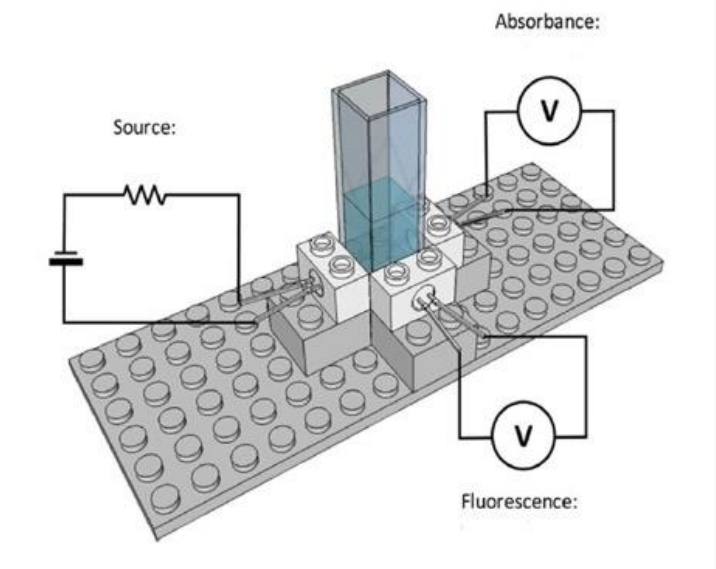


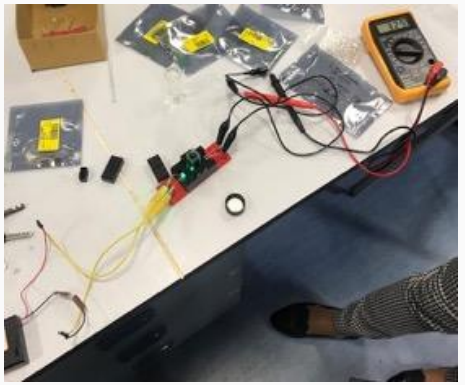
- Reflection – verify incidence/reflection
- Refraction – refractive index
- Scattering – single and multiple
- Diffraction – measuring the wavelength of light

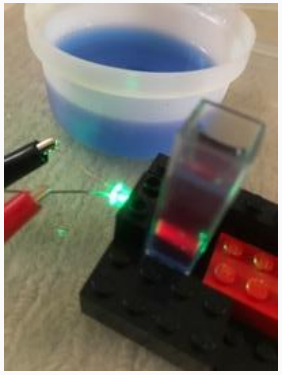
Introduction to spectroscopy – *LED Colorimeter*

- Using LEDs a photodiode and some Lego to make a colorimeter and measure absorbances of different materials.
- Test and verify the Beer-Lambert Law

$$A = \epsilon c l$$





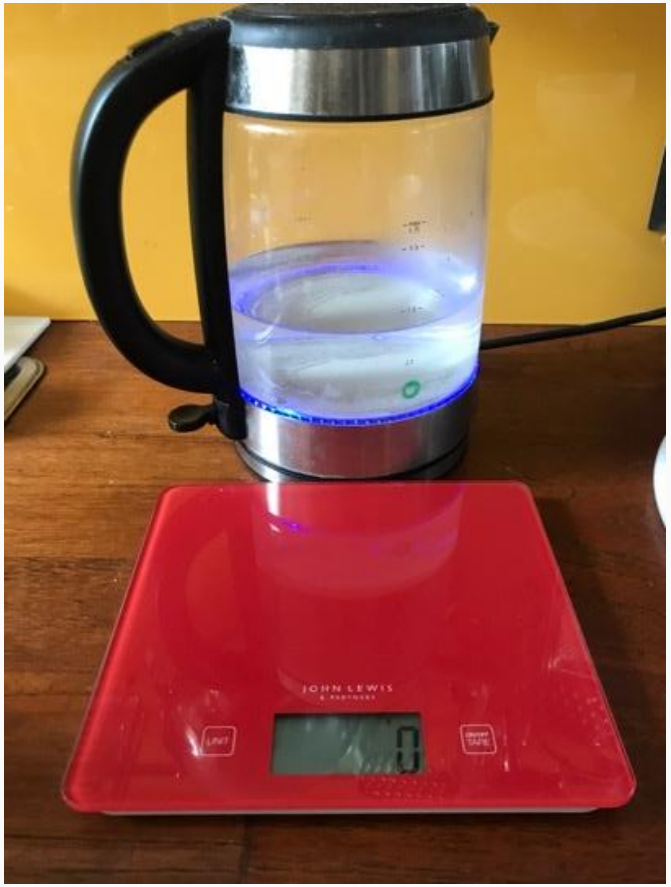


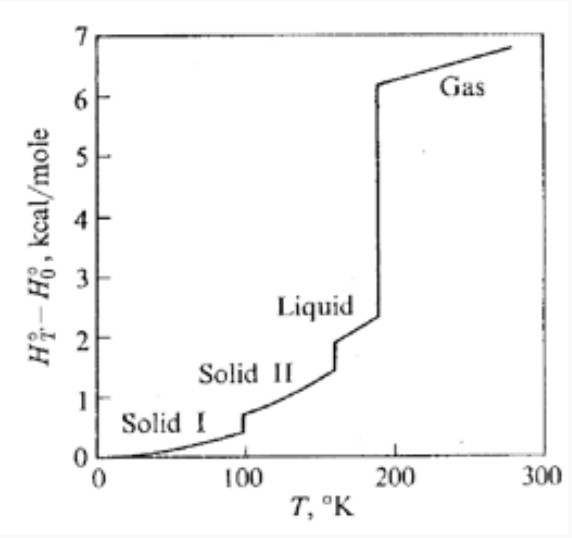
Estimating your Carbon Footprint



- Quantify your environmental impact (e.g. travel / heating / energy use).
- Use information you can find yourself (e.g., electricity/gas use from your energy meter; distance travelled from Google maps) and online tools to generate real data.
- Track how this changes over time; think about how it can be analysed/presented.

The Kettle as a Calorimeter



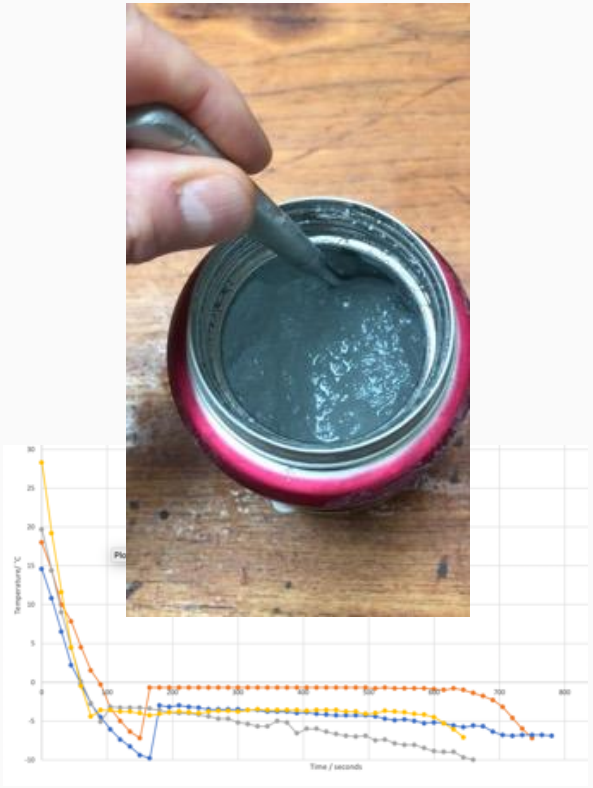




- Measure thermodynamic properties of water like the specific heat capacity and the enthalpy of vaporisation.

Cooling curves and colligative properties





- Why does adding salt to water lower the freezing point of the solution?
- You'll perform some experiments to measure this.
- Adding salt to ice lowers the temperature - how can we use this?

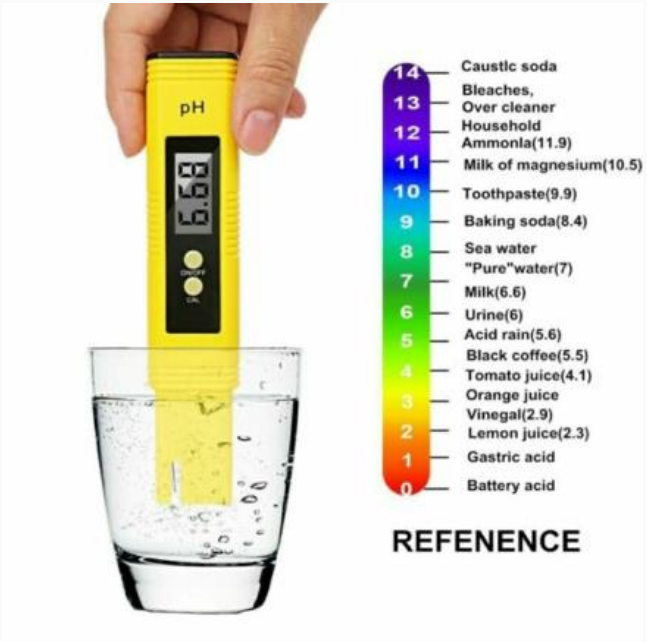
Making ice cream

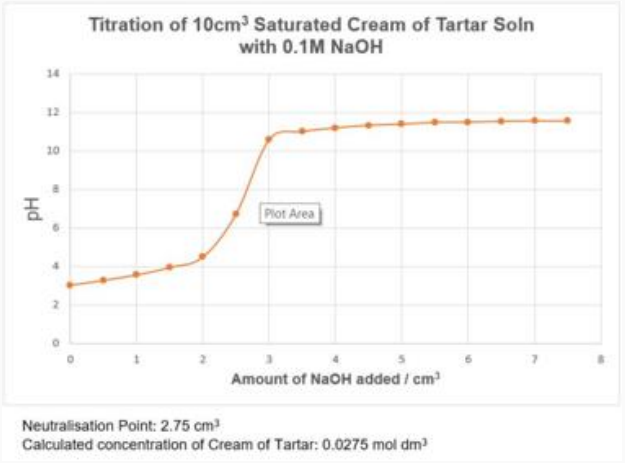






Acids and Bases – using a pH meter



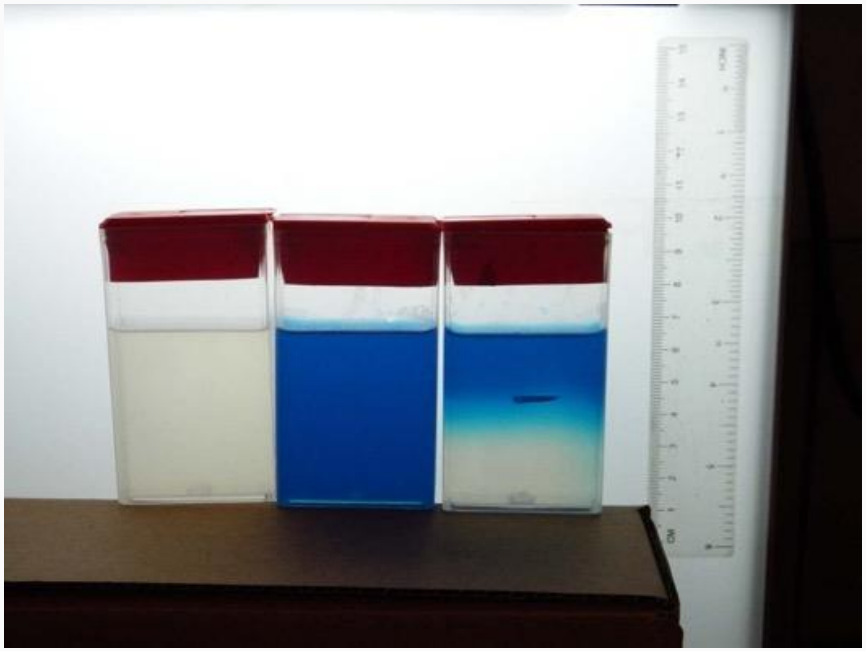


- Titration of vinegar
- Titration cream of tartar
- Titration of citric acid

- Measure the pH of everything you can find
- Measure the pH of every bodily fluid
- Measure the pH of puddles in the street

Measuring Diffusion Constants

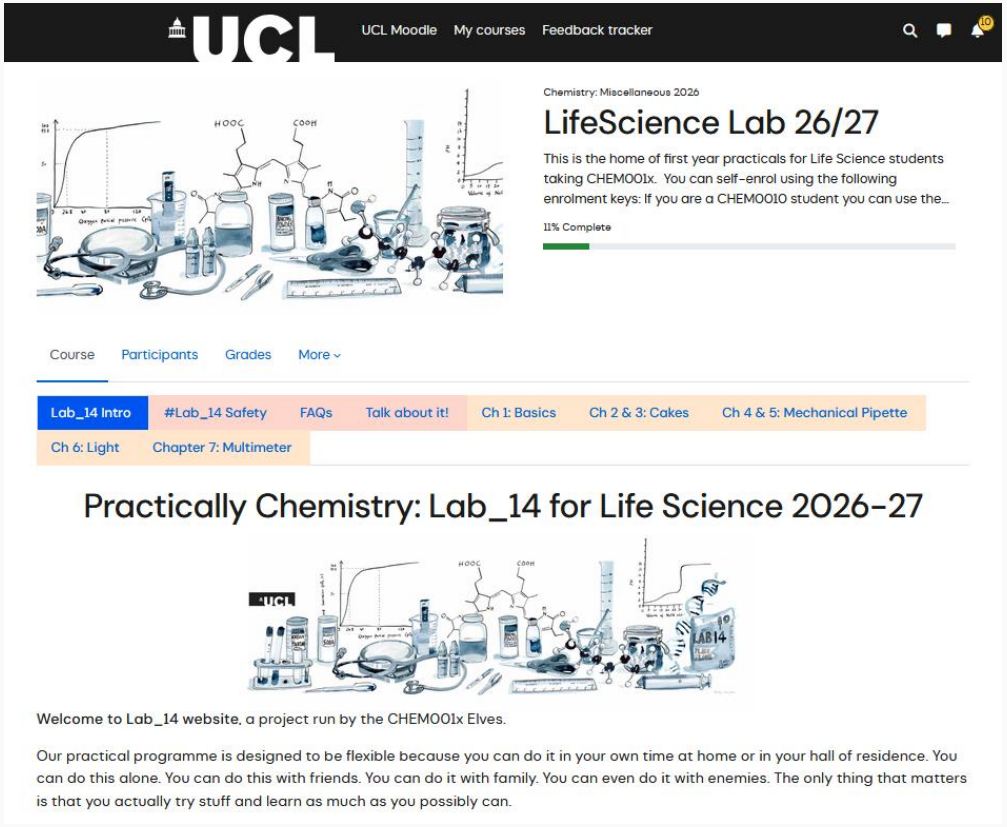
- Diffusion is avital process in chemistry and biology
- Use TicTac boxes as cell holders to measure rates of diffusion of food colouring through a viscous medium.
- Using the camera on your phone as a scientific measuring device.
- Using Image analysis software (ImageJ) for analysing results.



Lab 14 – How do I get help?

- We have a dedicated team of ‘Lab 14 elves’ ready to help you.
- **General enquiries** – chem001xlab@ucl.ac.uk
- **Online support** - The Lab_14 Forum on Moodle will run an online helpdesk on Monday, Tuesday, Thursday and Friday each week of term 1 and 2. Staff or PGTAs will be online to answer any questions.
- **In-person support** – Each week of Terms 1 and 2 (apart from reading week) there will be an in-person helpdesk in the Chemistry Department where you can come along to ask any questions about the practicals and to get replacements for missing or broken lab kit.
- The in-person sessions are in the same place where you will collect your lab kits from (room 289 in the Christopher Ingold Building)

The Lab 14 Moodle page



And finally, the most important piece of advice for Lab 14 – have some fun! (and cake)









